Hospital Patient Clinical Details

1. Project Overview and Objective

Overview

This project focuses on the advanced analysis of inpatient clinical records across a cohort of unique patient tracks and specific clinical encounters (`ENC-XXXXX`). The foundational dataset captures complete diagnostic, procedural, vital sign, and laboratory profiles for patients admitted under various clinical circumstances. The scope of work involves cleaning, transforming, and analysing raw data using Excel, and subsequently building an interactive Power BI dashboard to derive meaningful, actionable insights.

Objective

The main objective is to cleanse, standardize, and enhance the dataset across all clinical encounters (`ENC-XXXXX`). This includes addressing missing data points (such as unrecorded SpO_2 levels), eliminating inconsistencies, and normalizing data fields. Additionally, the project aims to derive advanced clinical metrics by classifying categorical groups.

.2. Data Sources

- **Source Description and Timeline:** **Kaggle.com and** 2022-2023.
- **Domain**: Health care Data set Analytics

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3. Problem Statement

- To analyse the Healthcare records to extract patient details.
- Focus on chronic condition and hypertension management. safety.
- Evaluate Hospital record to improve clinical decision-making.
- Focus on clinical risk stratification &patient safety.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Patient _id	Numeric (Integer)	Unique identifier for each Patient
Encounter _id	Numeric (Integer)	Unique identifier for each Encounter
Clinical-id	Numeric (Integer)	Unique identifier for Clinical department
Primary Diagnosis Code	Categorical	Standardized medical classification code

Primary diagnosis name	Categorical	The official text description corresponding to the primary diagnosis code
Disease severity	Categorical	The clinical grade or classification level of the condition (e.g., Mild, Moderate, Severe)
Primary procedure name	Categorical	The text name or standard classification of the main medical procedure performed
Procedure complexity	Categorical	The graded difficulty level of the procedure (e.g., Minor, Major, Complex)
Hospital acquired infection	Numeric (Integer)	A binary indicator flag (e.g., 0 = No, 1 = Yes) showing if an infection developed during the hospital stay
Blood glucose mg dl	Numeric (Integer)	The patient's blood sugar level measured in milligrams per deciliter
Serum creatinine mg dl	Numeric (Integer)	A lab measurement of kidney function tracking creatinine levels in milligrams per deciliter
Hemoglobin g dl	Numeric (Integer)	A lab test measuring the amount of oxygen-carrying protein in red blood cells in grams per deciliter
White blood cell count	Numeric (Integer)	The concentration of immune cells in the blood, typically measured to check for infection or inflammation
Systolic bp mm hg	Numeric (Integer)	The top number in a blood pressure reading, measuring the pressure in arteries during a heartbeat in millimeters of mercury
Diastolic bp mm hg	Numeric (Integer)	The bottom number in a blood pressure reading, measuring artery pressure between heartbeats
Heart rate bpm	Numeric (Integer)	The number of times the heart beats per minute

Respiratory rate bpm	Numeric (Integer)	The number of breaths a person takes per minute
Temperature Celsius	Numeric (Integer)	The core body temperature of the patient measured in degrees Celsius
spo2_percent	Numeric (Integer)	The oxygen saturation level in the blood, representing the percentage of oxygen-carrying hemoglobin

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5. Tools & Technologies

- **Excel:** Data cleaning, Missing, Values transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, Relationship Table and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query)

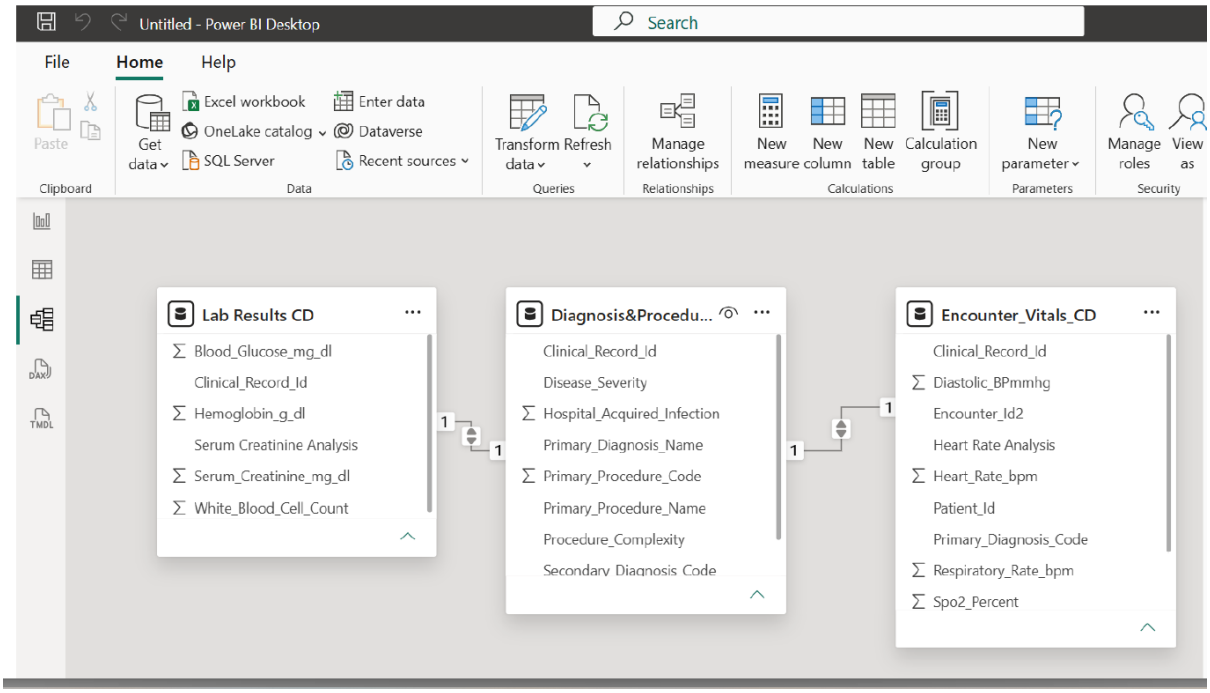
Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

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7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.



SCREENSHOT OF DATA MODELLING

Calculated Columns : Implemented DAX formulas to calculate important performance indicators and analyze patient records.

- **DAX Measures:** Implemented DAX formulas for key metrics,
- High WBC count=Calculate ()
- Total Hospital Infection=Sum ()
- Patient Count=DISTINCTCOUNT()
- Spo2 Value = SUM ()
- Total Clinical Record = COUNT ()
- Total Encounters = COUNT ()
- Average Glucose = AVERAGE ()

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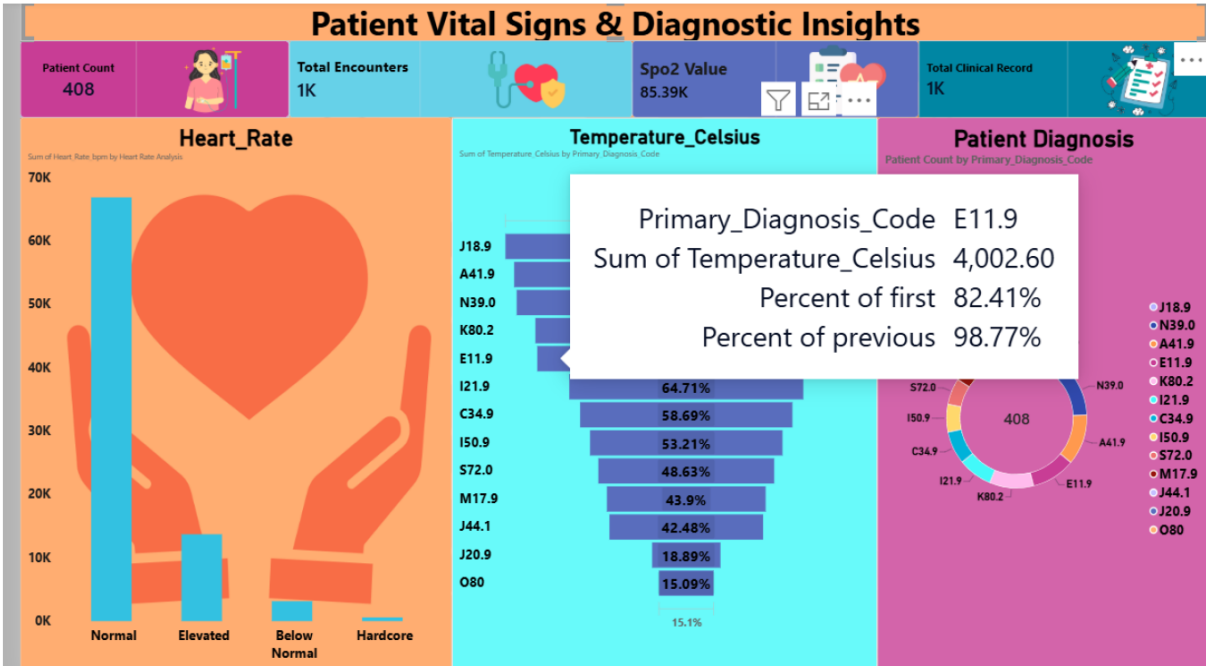
8. Analysis and Visualizations (Power BI)

Dashboard Features: Created interactive Power BI dashboards to analyse the Hospital Records and Patient details.

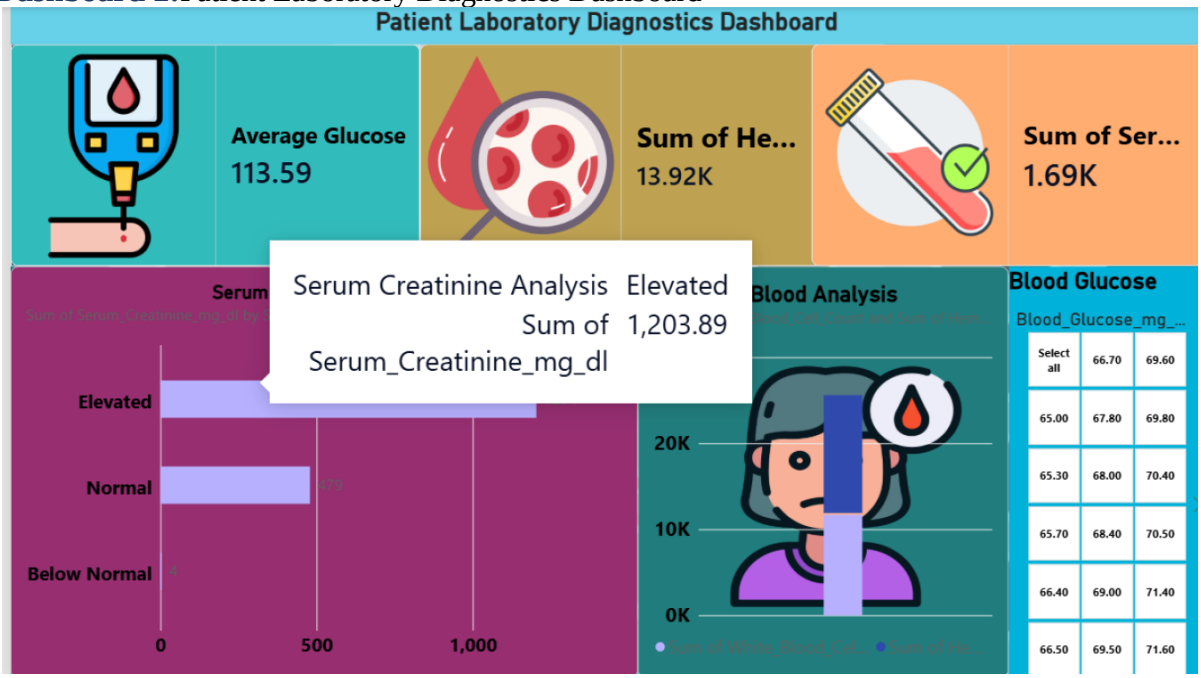
Card Visuals =Total Patient count, Total Encounters, Total Spo2 Value, Total Clinical Record, Average Glucose, Sum of Heart rate, Sum of serum Creatinine, sum of hospital infection,
Donut Chart=Visualize patient count by patient diagnosis code
Pie Chart=Compare disease severity range
Clustered Column Chart=Compare heart rate details and Heart rate Analysis
Funnel Chart=Sum of Temperature by primary Diagnosis code
Table=Primary Diagnosis and count of primary diagnosis code
Stacked Bar=Compare Sum of Serum creatinine by Serum creatinine Analysis
stacked column chart=Total white blood cell count and sum of hemoglobin analysis
slicer=used Blood glucose level in patient
Tile-styled button slicer= Utilized a text search slicer on the secondary diagnosis code field to allow users to efficiently look up specific clinical codes.

Visualizations (Power BI)

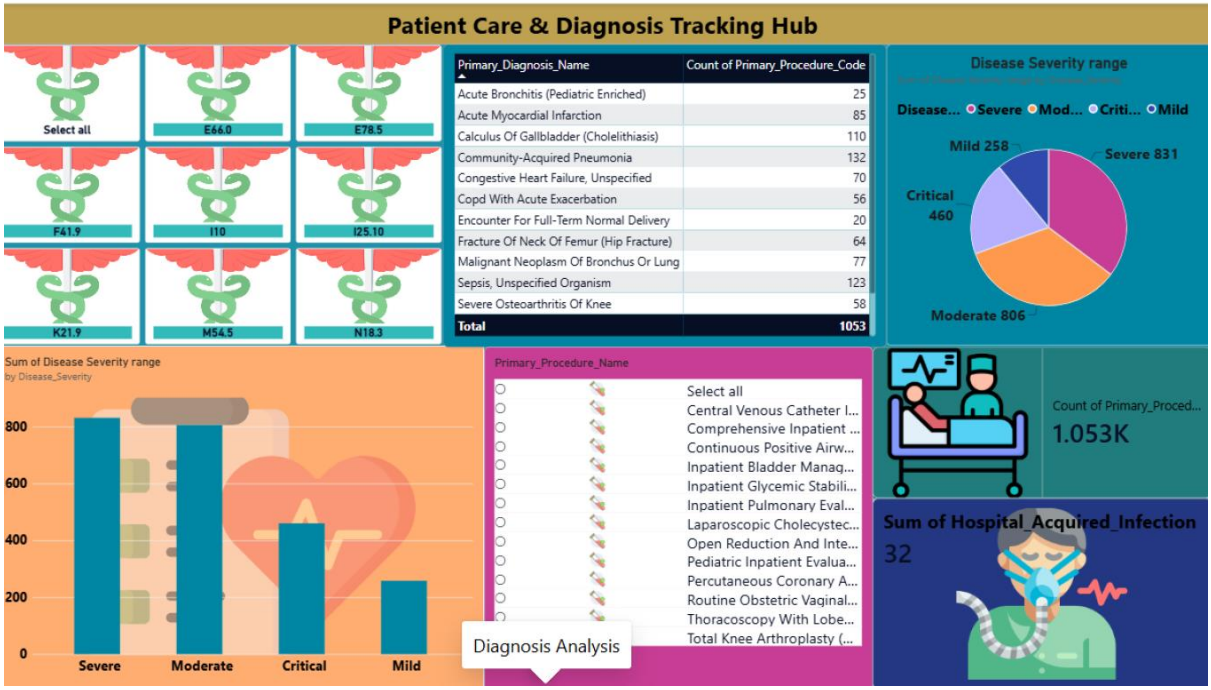
Dashboard 1: Patient Vital Signs & Diagnostic Insights



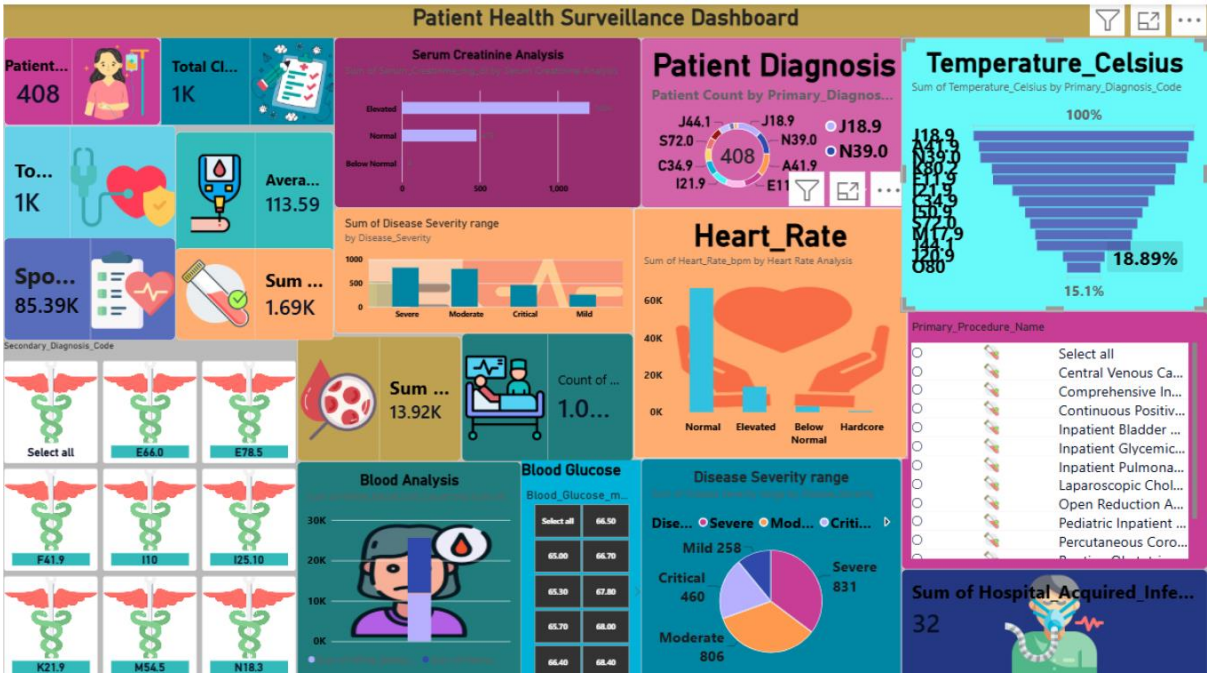
Dashboard 2: Patient Laboratory Diagnostics Dashboard



Dashboard 3: Patient Care & Diagnosis Tracking Hub Dashboard



Dashboard 4: Patient Laboratory Diagnostics Dashboard



SCREENSHOT OF DASHBOARD /REPORT

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9. Insights & Conclusions

- **Key Findings:** Summarize trends, patterns, or anomalies identified in the data
- **Provide the analysis insights:**
  - **Descriptive:** Synthesizes historical patient data to establish baseline trends, volumes, and patient health profiles.
  - **Diagnosis:** Drills down into data anomalies and correlations to uncover the root causes of clinical outcomes or operational bottlenecks.

- **Predictive:** Applies historical patterns and trends to forecast future patient risks, volumes, or clinical complications.
- **Prescriptive:** Leverages data insights to recommend specific clinical or operational actions to mitigate risks and improve outcomes

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10. Conclusions

This project successfully demonstrates how raw, siloed inpatient clinical records can be transformed into a unified, high-impact healthcare analytics solution. By leveraging the combined strengths of for Excel for data engineering and Power BI for interactive dashboard Creation.